Zad. 1

Wzory: $P(A|B) = \frac{P(A\cap B)}{P(B)}$,

$$P(A\cup B\cup C) = P(A)+P(B)+P(C)-P(A\cap B)-P(A\cap C)-P(B\cap C)+P(A\cap B\cap C)$$

$$P[(A\cap B\cap C)\,|\,(A\cap B)\cup(A\cap C)\cup(B\cap C)] =$$

$$= \frac{P[(A\cap B\cap C)\cap[(A\cap B)\cup(A\cap C)\cup(B\cap C)]]}{P[(A\cap B)\cup(A\cap C)\cup(B\cap C)]} =$$

$$= \frac{P[(A\cap B\cap C)\cup(A\cap B\cap C)\cup(A\cap B\cap C)]}{P(A\cap B)+P(A\cap C)+P(B\cap C)-3P(A\cap B\cap C)+P(A\cap B\cap C)} =$$

$$= \frac{P(A\cap B\cap C)}{P(A\cap B)+P(A\cap C)+P(B\cap C)-2P(A\cap B\cap C)} =$$

$$P(A|B\cap C) = \frac{P(A\cap B\cap C)}{P(B\cap C)}$$

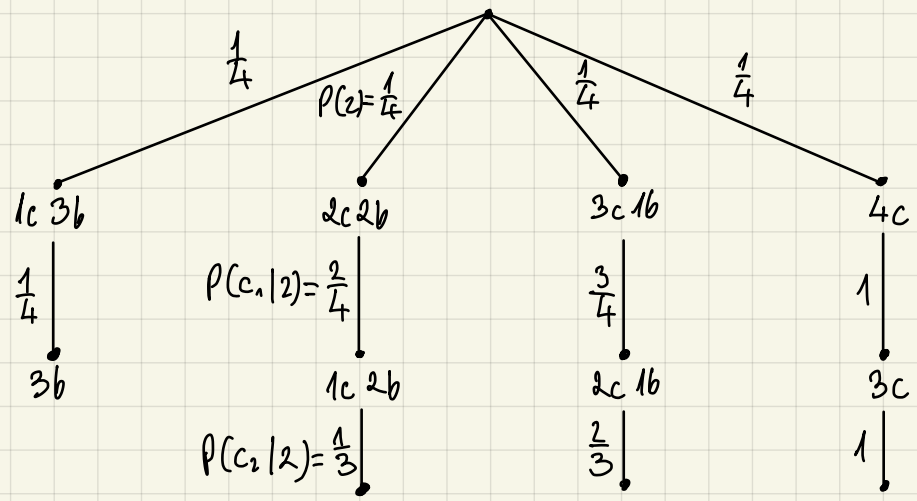
$$P(A\cap B\cap C) = P(A|B\cap C)P(B\cap C) = P(B|A\cap C)P(A\cap C) = P(C|A\cap B)P(A\cap B) =$$

$$= 0{,}6P(A\cap C) = 0{,}3P(A\cap C) = 0{,}9P(A\cap B)$$

$$= \frac{1}{\frac{1}{0{,}9}+\frac{1}{0{,}3}+\frac{1}{0{,}6}-2} = \frac{9}{37}$$

Zad. 2

Wzory: $P(A|B) = \sum_{k=1}^{n} P(A|C_k)P(C_k|B)$; $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$



c_1 - czarna kula w pierwszym wyciągnięciu

c_2 - czarna kula w drugim wyciągnięciu

k - numer urny

$$P(c_2|c_1) = \sum_{k=1}^{4} P(c_2|k)P(k|c_1)$$

$P(c_2|1) = 0$

$P(c_2|2) = \frac{1}{3}$

$P(c_3|3) = \frac{2}{3}$

$P(c_4|4) = 1$

$$P(k|c_1) = \frac{P(c_1|k)P(k)}{P(c_1)}$$

$$P(c_1) = \sum_{k=1}^{4} P(c_1|k)P(k) = \frac{1}{4}\cdot\frac{1}{4} + \frac{2}{4}\cdot\frac{1}{4} + \frac{3}{4}\cdot\frac{1}{4} + \frac{1}{4} = \frac{5}{8}$$

$$P(1|c_1) = \frac{\frac{1}{4}\cdot\frac{1}{4}}{\frac{5}{8}} = \frac{1}{10}$$

$P(2|c_1) = \frac{1}{5}$

$P(3|c_1) = \frac{3}{10}$

$P(4|c_1) = \frac{2}{5}$

$$P(c_2) = \frac{1}{3}\cdot\frac{1}{5} + \frac{2}{3}\cdot\frac{3}{10} + 1\cdot\frac{2}{5} = \frac{2}{3}$$

Zad. 3

Mediana: $P(X \leq m) \geq \frac{1}{2}$ i $P(X \geq m) \geq \frac{1}{2}$

Dystrybuanta i-tej statystyki porządkowej: $F_{X_i}(x) = \sum_{k=i}^{n} \binom{n}{k} [F_X(x)]^k [1 - F_X(x)]^{n-k}$

$P(X \leq m) = F_X(m) = \frac{1}{2}$

$P(X \geq m) = 1 - F_X(m) = \frac{1}{2}$

$$P(X_2 \le m \le X_7) = P(X_2 \le m) - P(X_7 \le m) = F_{X_2}(m) - F_{X_7}(m)$$

$$F_{X_{(i)}} = \sum_{k=i}^{8} \binom{8}{k} [F_X(m)]^k [1 - F_X(m)]^{8-k} = \sum_{k=i}^{8} \binom{8}{k} \left(\frac{1}{2}\right)^8$$

$$P(X_2 \le m \le X_7) = \left(\frac{1}{2}\right)^8 \left[\binom{8}{2} + \binom{8}{3} + \binom{8}{4} + \binom{8}{5} + \binom{8}{6} + \binom{8}{7} + \binom{8}{8} - \binom{8}{7} - \binom{8}{8}\right] = \frac{119}{128}$$

Zad. 4

Wzór: $P(X > x \mid Y > y) = \dfrac{P(X > x, Y > y)}{P(Y > y)}$

$$f(y) = \int_0^1 \frac{3}{4}x + 2xy + \frac{1}{4}y \, dx = \frac{3}{4} \cdot \frac{x^2}{2} + 2y\frac{x^2}{2} + \frac{1}{4}yx \Big|_0^1 =$$

$$= \frac{3}{8} + y + \frac{1}{4}y = \frac{5}{4}y + \frac{3}{8}$$

$$P\left(X > \frac{1}{2}, Y > \frac{1}{2}\right) = \int_{\frac{1}{2}}^{1} \int_{\frac{1}{2}}^{1} \frac{3}{4}x + 2xy + \frac{1}{4}y \, dx\, dy = \int_{\frac{1}{2}}^{1} \frac{3}{8}x^2 + yx^2 + \frac{1}{4}yx \Big|_{\frac{1}{2}}^{1} dy =$$

$$= \int_{\frac{1}{2}}^{1} \frac{3}{8} + y + \frac{1}{4}y - \frac{3}{32} - \frac{1}{4}y - \frac{1}{8}y \, dy = \int_{\frac{1}{2}}^{1} \frac{7}{8}y + \frac{9}{32} \, dy = \frac{15}{32}$$

$$P\left(Y > \frac{1}{2}\right) = \int_{\frac{1}{2}}^{1} \frac{5}{4}y + \frac{3}{8} \, dy = \frac{21}{32}$$

$$P\left(X > \frac{1}{2} \mid Y > \frac{1}{2}\right) = \frac{15}{32} \cdot \frac{32}{21} = \frac{5}{7}$$

Zad. 5

Wzór: $Var(X) = Var(E[X|Y]) + E[Var(X|Y)]$

Wartości oczekiwane w złożonym rozkładzie Poissona: $EX = \lambda EY$

Wariancja w złożonym rozkładzie Poissona: $Var(X) = \lambda(Var(Y) + (EY)^2)$

$Var(S) = Var(E[S|\Lambda]) + E[Var(S|\Lambda)]$

$E[S|\Lambda] = 2\lambda$

$Var(S|\Lambda) = \lambda \cdot (4+4) = 8\lambda$

$Var(S) = Var(2\Lambda) + E[8\Lambda] = 4\,Var(\Lambda) + 8E\Lambda$

$$EA = 1 \cdot \frac{3}{4} + 2 \cdot \frac{1}{4} = \frac{3}{4} + \frac{2}{4} = \frac{5}{4}$$

$$EA^2 = 1 \cdot \frac{3}{4} + 4 \cdot \frac{1}{4} = \frac{3}{4} + \frac{4}{4} = \frac{7}{4}$$

$$Var(A) = \frac{7}{4} - \frac{25}{16} = \frac{3}{16}$$

$$Var(S) = 4 \cdot \frac{3}{16} + 8 \cdot \frac{5}{4} = \frac{3}{4} + 10 = 10\frac{3}{4}$$

Zad. 6

$$S(c) = c \cdot \sum_{i=1}^{n} (x_i - \bar{x})^2$$

$$MSE(S) = Var(S) + (B(S))^2$$

$$S(c) = c\sigma^2 \underbrace{\frac{1}{\sigma^2} \sum_{i=1}^{n} (x_i - \bar{x})^2}_{\chi^2(n-1)}$$

$$Var(S) = Var(c\sigma^2 X) = c^2\sigma^4 Var(X) = c^2\sigma^4 2(n-1)$$

$$E(S) = E(c\sigma^2 X) = c\sigma^2 E(X) = c\sigma^2 (n-1)$$

$$B(S) = E(S) - \sigma^2 = c\sigma^2(n-1) - \sigma^2 = \sigma^2(cn - c - 1)$$

$$(B(S))^2 = \sigma^4 (cn - c - 1)^2 \qquad (cn - c - 1)(cn - c - 1)$$

$$MSE(S) = 2c^2\sigma^4(n-1) + \sigma^4(cn - c - 1)^2 =$$

$$= 2c^2\sigma^4(n-1) + \sigma^4(c^2n^2 + c^2 + 1 - 2c^2n - 2cn + 2c)$$

$$MSE'(S) = 4c\sigma^4(n-1) + \sigma^4(2cn^2 + 2c - 4cn - 2n + 2) = 0 \quad /:\sigma^4$$

$$4c(n-1) + 2cn^2 + 2c - 4cn = 2n - 2 \quad /:2$$

$$2c(n-1) + cn^2 + c - 2cn = n - 1$$

$$c(\cancel{2n} - 2 + n^2 + 1 - \cancel{2n}) = n - 1$$

$$c(n^2 - 1) = n - 1$$

$$c = \frac{n-1}{n^2-1} = \frac{n-1}{(n-1)(n+1)} = \frac{1}{n+1}$$

Zad. 7

$X_1, \dots, X_n \sim U(0, \varphi)$

$$Var\left(\frac{n+1}{n} \max\{X_1, \dots, X_n\}\right) = \frac{(n+1)^2}{n^2} Var(\max\{X_1, \dots, X_n\})$$

$$F_X(x) = \frac{x}{\varphi}$$

$$F_{max}(x) = \left(\frac{x}{\varphi}\right)^n$$

$$E(max) = \int_0^\varphi 1 - \left(\frac{x}{\varphi}\right)^n dx = x - \frac{x^{n+1}}{(n+1)\varphi^n}\Bigg|_0^\varphi =$$

$$= \varphi - \frac{\varphi^{n+1}}{(n+1)\varphi^n} = \frac{\varphi^{n+1}(n+1) - \varphi^{n+1}}{(n+1)\varphi^n} = \frac{n\varphi^{n+1}}{(n+1)\varphi^n} = \frac{n\varphi}{n+1}$$

$$E(max^2) = \int_0^\varphi 2x - 2x\left(\frac{x}{\varphi}\right)^n dx = \int_0^\varphi 2x - \frac{2}{\varphi^n} x^{n+1} dx =$$

$$= \frac{2x^2}{2} - \frac{2}{\varphi^n} \frac{x^{n+2}}{n+2}\Bigg|_0^\varphi = \varphi^2 - \frac{2\varphi^{n+2}}{\varphi^n(n+2)} = \frac{\varphi^{n+2}(n+2) - 2\varphi^{n+1}}{(n+2)\varphi^n} =$$

$$= \frac{n\varphi^{n+2}}{(n+2)\varphi^n} = \frac{n\varphi^2}{n+2}$$

$$Var(max) = \frac{n\varphi^2}{n+2} - \frac{n^2\varphi^2}{(n+1)^2} = \frac{n\varphi^2(n+1)^2 - n^2\varphi^2(n+2)}{(n+2)(n+1)^2} =$$

$$= \frac{n\varphi^2(n^2+2n+1) - n^2\varphi^2(n+2)}{(n+2)(n+1)^2} = \frac{\cancel{n^3\varphi^2} + \cancel{2n^2\varphi^2} + n\varphi^2 - \cancel{n^3\varphi^2} - \cancel{2n^2\varphi^2}}{(n+2)(n+1)^2} =$$

$$= \frac{n\varphi^2}{(n+2)(n+1)^2}$$

$$Var = \frac{\cancel{(n+1)^2}}{n^2} \cdot \frac{n\varphi^2}{(n+2)\cancel{(n+1)^2}} = \frac{\varphi^2}{n(n+2)}$$

Zad. 2

TW. Rozważmy stacjonarny łańcuch Markowa $\{X_n, n = 0, 1, 2, \dots\}$, gdzie $X_n \in S = \{0, 1, 2, \dots, r\}$. Załóżmy, że łańcuch jest nieredukowalny i aperiodyczny. Wtedy:

1. Układ równań

$$\begin{cases} \pi = \pi P \\ \sum_{j \in S} \pi_j = 1 \end{cases}$$

ma jedno rozwiązanie.

2. Rozwiązanie powyższego układu jest rozkładem granicznym łańcucha Markowa czyli:

$$\pi_j = \lim_{n \to \infty} P(X_n = j \mid X_0 = i) \quad \forall\ i, j \in S$$

Łańcuch Markowa w zadaniu jest nieredukowalny ponieważ można przejść z dowolnego stanu do innego dowolnego stanu. Jest aperiodyczny ponieważ można przejść np. ze stanu 1 do stanu 1 w jednym kroku.

$$\begin{bmatrix} \frac{1}{7} & \frac{2}{7} & \frac{4}{7} \end{bmatrix} \begin{bmatrix} q & p & 0 \\ q & 0 & p \\ 0 & q & p \end{bmatrix} = \begin{bmatrix} \frac{1}{7} & \frac{2}{7} & \frac{4}{7} \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{7}q + \frac{2}{7}q & \frac{1}{7}p + \frac{4}{7}q & \frac{2}{7}p + \frac{4}{7}p \end{bmatrix} = \begin{bmatrix} \frac{1}{7} & \frac{2}{7} & \frac{4}{7} \end{bmatrix}$$

$$\begin{bmatrix} \frac{3}{7}q & \frac{1}{7}p + \frac{4}{7}q & \frac{6}{7}p \end{bmatrix} = \begin{bmatrix} \frac{1}{7} & \frac{2}{7} & \frac{4}{7} \end{bmatrix}$$

$$\frac{3}{7}q = \frac{1}{7}$$

$$q = \frac{1}{3}$$

Zad. 9

$X_1, \dots, X_n \sim P(\lambda)$

$\hat{p}_0 = e^{-\bar{X}}$

$$E(\hat{p}_0) = E(e^{-\bar{X}}) = E\left[e^{-\frac{1}{n}(X_1 + \dots + X_n)}\right] = \left[E\left(e^{-\frac{X}{n}}\right)\right]^n =$$

$$= \left[M_X\left(-\frac{1}{n}\right)\right]^n = \left[\exp\left(\lambda\left(e^{-\frac{1}{n}} - 1\right)\right)\right]^n = \exp\left(n\lambda\left(e^{-\frac{1}{n}} - 1\right)\right)$$

$$B(\hat{p}_0) = E(\hat{p}_0) - p_0 = \exp\left(n\lambda\left(e^{-\frac{1}{n}} - 1\right)\right) - \exp(-\lambda) \overset{?}{>=<} 0 \quad / \ln$$

$$n\lambda\left(e^{-\frac{1}{n}} - 1\right) + \lambda \overset{?}{>=<} 0 \quad / : \lambda$$

$$n\left(e^{-\frac{1}{n}} - 1\right) + 1 \overset{?}{>=<} 0$$

$$f(n) = n e^{-\frac{1}{n}} - n + 1$$

$$f'(n) = n e^{-\frac{1}{n}} \cdot \frac{1}{n^2} + e^{-\frac{1}{n}} - 1 = \frac{1}{n} e^{-\frac{1}{n}} + e^{-\frac{1}{n}} - 1$$

$$f''(n) = \frac{1}{n^2} e^{-\frac{1}{n}} - \frac{1}{n^2} e^{-\frac{1}{n}} + \frac{1}{n^3} e^{-\frac{1}{n}} = \frac{1}{n^3} e^{-\frac{1}{2}} > 0 \Rightarrow f'(n) \text{ rośnie}$$

$\lim_{n \to \infty} f'(n) = 0$ czyli $f'(n) < 0$ stąd $f(n)$ maleje

$f(1) = 0{,}6321$

$$\lim_{n \to \infty} f(n) = \lim_{n \to \infty} n\left(e^{-\frac{1}{n}} - 1\right) + 1 = \lim_{n \to \infty} \frac{e^{-\frac{1}{n}} - 1}{\frac{1}{n}} + 1 = \lim_{n \to \infty} \frac{\frac{1}{n^2} e^{-\frac{1}{n}}}{-\frac{1}{n^2}} + 1 =$$

$$= \lim_{n \to \infty} -e^{-\frac{1}{n}} + 1 = 0$$

$f(n) > 0$

Obciążenie dodatnie

Zad. 10

$H_0 : a = 1 \qquad f_0(x) = 2x$

$H_1 : a = 2 \qquad f_1(x) = 3x^2$

Test ma być najmocniejszy więc można zastosować likelihood ratio test.

Odrzucamy H_0 gdy:

$$\lambda(x) = \frac{L(x; a_1)}{L(x; a_0)} = \frac{3x^2}{2x} = \frac{3}{2}x > c$$

$X > c$

Obszar krytyczny:

$K = \{x : x > c\}$

$P_0(K) = \alpha, \quad P_1(K) = 1 - \beta$

$P(X > c \mid H_0) = 0,1$

$$P(X > c \mid H_0) = \int_c^1 2x \, dx = x^2 \Big|_c^1 = 1 - c^2 = 0,1$$

$c = \sqrt{0,9}$